

[Dash...](#) / [My co...](#) / [MA-224...](#) / [I...](#) / [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of i...](#)

Status	Finished
Started	Thursday, 6 November 2025, 12:00 PM
Completed	Thursday, 6 November 2025, 12:13 PM
Duration	12 mins 37 secs
Marks	1.33/3.00
Grade	1.33 out of 3.00 (44.45%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.33 out of 1.00

Consider the recurrence relation

$$4 \cdot a_n - 8 \cdot a_{n-1} + 8 \cdot a_{n-2} = 0.$$

Let $\{a_n\}_n$ and $\{b_n\}_n$ be two numerical sequences. We say that a_n is a *solution* if the sequence $\{a_n\}_n$ satisfies the recurrence relation, and likewise for b_n .

Select every statement below which is true, and leave the false statements unchecked.

- ☒ a_n and b_n are solutions, and if $a_1=b_1$, then $a_n=b_n$ for each $n \geq 1$.
- ☐ If $a_0=0$, $a_1=-128$, and $a_2=-253$, then a_n could be a solution.
- ☒ If a_n is a solution, and $a_1=a_2=0$, then $a_n=0$ for each $n \geq 1$.
- ☐ If a_n is a solution, and $a_0=2$ and $a_1=-2$, then a_n is an integer for each $n \geq 1$.
- ☐ If $a_0=-2$, $a_1=1$, then a_n can not be a solution.
- ☒ If a_n is a solution, then $b_n=18a_n$ is also a solution.
- ☐ None of the above

Question **2**

Incorrect

Mark 0.00 out of 1.00

In this problem you are tasked with computing subgroups of $(\mathbb{Z}/n, +_n)$ and direct products of such groups.

a) Find the subgroup generated by the element $2 \in \mathbb{Z}/8$.

$\langle 2 \rangle =$

Your last answer was interpreted as follows:

$$\left\{ \frac{1}{8}, \frac{2}{8} \right\}$$

Syntax guide: Input your answer as a set. E.g. if you think the answer is $\{1, 2, 3\}$, then you should type **{1,2,3}**.

b) Find the subgroup generated by the element $2 \in \mathbb{Z}/6$.

$\langle 2 \rangle =$

Your last answer was interpreted as follows:

$$\left\{ \frac{1}{6}, \frac{2}{6} \right\}$$

Syntax guide: Input your answer as a set. E.g. if you think the answer is $\{1, 2, 3\}$, then you should type **{1,2,3}**.

c) Find the subgroup generated by the element $[2, 2] \in \mathbb{Z}/8 \times \mathbb{Z}/6$.

$\langle [2, 2] \rangle =$

Syntax guide: Elements of this group are on the form $[a, b]$. Your answer should be a set of such elements. E.g. if you think the answer is $\{[1, 2], [0, 2], [1, 3]\}$, then you should type **{[1,2],[0,2],[1,3]}**.

Question **3**

Correct

Mark 1.00 out of 1.00

a) Let $G = (\mathbb{Z}/2)^{\times 4}$, and let $H \subset G$ be the subgroup

$$H = \{[0,0,0,0], [0,0,0,1], [0,0,1,0], [0,0,1,1], [0,1,0,0], [0,1,0,1], [0,1,1,0], [0,1,1,1]\}.$$

What is the index of $H \subset G$?

$[G : H] =$

Your last answer was interpreted as follows:

2

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that rotates 180 degrees around the origin.

Compute the index of the subgroup generated by σ :

$[D_4 : \langle \sigma \rangle] =$

Your last answer was interpreted as follows:

4

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 275

Signing code:

Your last answer was interpreted as follows:

54892

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

Jump to...